BEFORE YOU READ

Where vector spaces show up

Vector spaces are not only arrows: polynomials, matrices and functions form vector spaces too, and many familiar sets sit inside them as subspaces.

- L0 **Common vector spaces**
  - L1 **Finite-dimensional**
    - L2 $\mathbb{R}^n$ n-tuples of real numbers
    - L2 $\mathbf{P_n}$ polynomials of degree at most n — $\dim n + 1$
    - L2 **m × n matrices** — $\dim mn$
  - L1 **Infinite-dimensional**
    - L2 **Continuous functions** on an interval
  - L1 **Subspace examples**
    - L2 **Solutions of a homogeneous system**
    - L2 **Symmetric n × n matrices**
    - L2 **Polynomials with p(0) = 0**
    - L2 **Solutions of y″ + y = 0** basis cos x, sin x

TAKEAWAY

Same axioms, very different objects

Anything that satisfies the axioms is a vector space, whatever its elements look like.